

Munjung Kim

Email: qns8tc@virginia.edu
Homepage: munjungkim.github.io

Education	University of Virginia <i>Doctor of Philosophy, Data Science</i> • Advisor: Dr. Yong-Yeol Ahn Indiana University Bloomington <i>Doctor of Philosophy program, Informatics (transferred with advisor)</i> • Advisor: Dr. Yong-Yeol Ahn Pohang University of Science and Technology <i>Bachelor of Science, Physics</i>	Aug 2025 – Aug 2027 (expected) Aug 2022 – Aug 2025 Feb 2017 – Feb 2022
Work Experience	Nokia Bell Labs Research Intern • Quantified U.S. occupational tasks' exposure to disruptive AI using LLMs, text embeddings, and patent citation-network analysis [A2] SK Hynix Quality Control Intern	<i>Cambridge, UK</i> June 2024 – Aug 2024 <i>Icheon, South Korea</i> Jan 2020 – Feb 2020
Research Summary	I study the emergence and societal impact of scientific and technological breakthroughs. Drawing on network science, machine learning, and large-scale text analysis, I develop systematic measures to identify them, understand how they emerge, and assess how they impact technology policy and labor. Through this approach, my work bridges advanced computational methods with innovation studies and technology policy.	
Research Interest	Innovation & Science of Science; Computational Social Science; AI Policy & The Future of Work; Network Science & Machine Learning	
Publications & Preprints	[A1] Munjung Kim, Sadamori Kojaku, and Yong-Yeol Ahn. "Uncovering simultaneous breakthroughs with a robust measure of disruptiveness". <i>Science Advances</i> 12 (2026). doi: 10.1126/sciadv.adx3420. [code] [pip] [blog] <i>Featured in Physics (APS), Phys.org, Yahoo News, among others. Invited as a panelist to the National Academies workshop on Disruptive Innovation (May 2026) based on this work.</i> [A2] Munjung Kim, Marios Constantinides, Sanja Šćepanović, Yong-Yeol Ahn, and Daniele Quercia. <i>Disruptive AI Increasingly Reaches Collaborative Work</i>. [arXiv:2507.11403]. 2025. <i>preprint</i>. [A3] Rachith Aiyappa, Xin Wang, Munjung Kim, Ozgur Seckin, Yong-Yeol Ahn, and Sadamori Kojaku. "Implicit Degree Bias in the Link Prediction Task". <i>Proceedings of the 42nd International Conference on Machine Learning (ICML)</i>. Honolulu, Hawaii, 2025. [A4] Munjung Kim, Tuğrulcan Elmas, and Filippo Menczer. "Toxic Synergy Between Hate Speech and Fake News Exposure". <i>Proceedings of the 18th International AAAI Conference on Web and Social Media (ICWSM)</i>. 2024. [A5] Matthew R DeVerna, Rachith Aiyappa, Wanying Zhao, Manita Pote, Bao Tran Truong, David Axelrod, Aria Pessianzadeh, Zoher Kachwala, Munjung Kim, Ozgur Can Seckin, et al. "A multi-platform collection of social media posts about the 2022 US midterm elections". <i>Proceedings of the 17th International AAAI Conference on Web and Social Media (ICWSM)</i>. 2023. [A6] Munjung Kim, Jisung Yoon, Woo-Sung Jung, and Hyunuk Kim. "Quantifying the topic disparity of scientific articles". <i>Companion Proceedings of the Web Conference</i>. 2022, pp. 769–773.	

Working Papers	[W1] Munjung Kim , Jisung Yoon, Yong-Yeol Ahn, and Jaehyuk Park. <i>AI Policy Discourse Diverges Along Development Lines</i> . 2026. <i>working paper; accepted to Research Policy ECR Conference 2026</i> .
	[W2] Munjung Kim , Nadav Kunievsky, Robert Ward, Seolmin Yang, and Yong-Yeol Ahn. <i>Murmuration of Scientists</i> . Models scientists' movement across research topics as collective flocking (boids model) driven by peer influence. 2026. <i>in preparation; presented at NetSci 2026</i> .
Oral Presentations	Invited Talks
	• <i>Panelist</i>, Understanding and Scoping Future Research on Disruptive Innovation, National Academies of Sciences, Engineering, and Medicine, Washington, DC 05/20/2026 • UVA SDS Advisory Board Meeting 04/17/2026 • KISTI 01/24/2025 • SNU Business School Workshop 01/03/2025 • KAIST STP Science of Science Research Seminar 12/18/2024 • KDI School Special Lecture 12/17/2024
	Contributed Talks
	Munjung Kim and Yong-Yeol Ahn. "Murmuration of Scientists". NetSci. 2026.
	Munjung Kim , Marios Constantinides, Sanja Šćepanović, Yong-Yeol Ahn, and Daniele Querchia. "Disruptive AI Increasingly Reaches Collaborative Work". IHS Graduate Conference. 2025.
	Munjung Kim , Sadamori Kojaku, and Yong-Yeol Ahn. "A Continuous Embedding Measure of Disruptiveness Reveals Hidden Simultaneous Disruption". IC2S2. 2024.
	Munjung Kim , Sadamori Kojaku, and Yong-Yeol Ahn. "Quantifying disruptiveness using neural embedding method". NetSci. 2023.
	Munjung Kim , Jisung Yoon, Hyunuk Kim, and Woo-Sung Jung. "Neural embedding of research papers reveals characteristics of policy research institutes". Networks: A Joint Sunbelt & NetSci Conference. 2021.
	Munjung Kim , Jisung Yoon, and Woo-Sung Jung. "Quantitative Analysis on Institutions' Activity Policy Research with Embedding Method". Korea Academy of Complexity Studies. 2019.
Honors & Awards	Humane Studies Fellowship , Institute for Humane Studies 2024
	Complex Systems Summer School , Santa Fe Institute 2023
	National Scholarship for Excellence (Sci. & Eng.) , Korea Student Aid Foundation 2017–2021
	Semester of High Honor , POSTECH 2019–2021
	Global Talent Attraction Program , Indiana University (deferred, COVID-19) 2020
	Senior Mentoring Program Scholarship , POSTECH 2019–2020
Teaching	Tutorial Instructor , KDI School AI Special Seminar 07/2026
	• "AI-Assisted Research" [materials]
	Tutorial Instructor , SICSS Korea, KAIST 06/2026
	• "Claude Code in Research for Beginners" (introductory tutorial) [materials]
	Senior Tutor , Senior Mentoring Program, POSTECH 2019–2020
	• Tutored Computational Physics and Calculus II
Skills	Computer Language : Python, R, MATLAB, C, C++, Cython
	Workflow for Reproducible Data Analysis : Snakemake
	Analysis Modules : PyTorch, scikit-learn, NumPy, SciPy, Pandas, igraph, NetworkX
	Languages : English (Proficient), Korean (Native)
Service	Journal Reviewer : <i>Nature</i> (Matters Arising), <i>Network Science</i> , <i>npj Complexity</i> 2025
	Conference Reviewer : International Conference on Computational Social Science (IC2S2, 2024–2025); International Society for Scientometrics and Informetrics (ISSI, 2023); International Workshop on Cyber Social Threats (CySoc, 2023–2024)